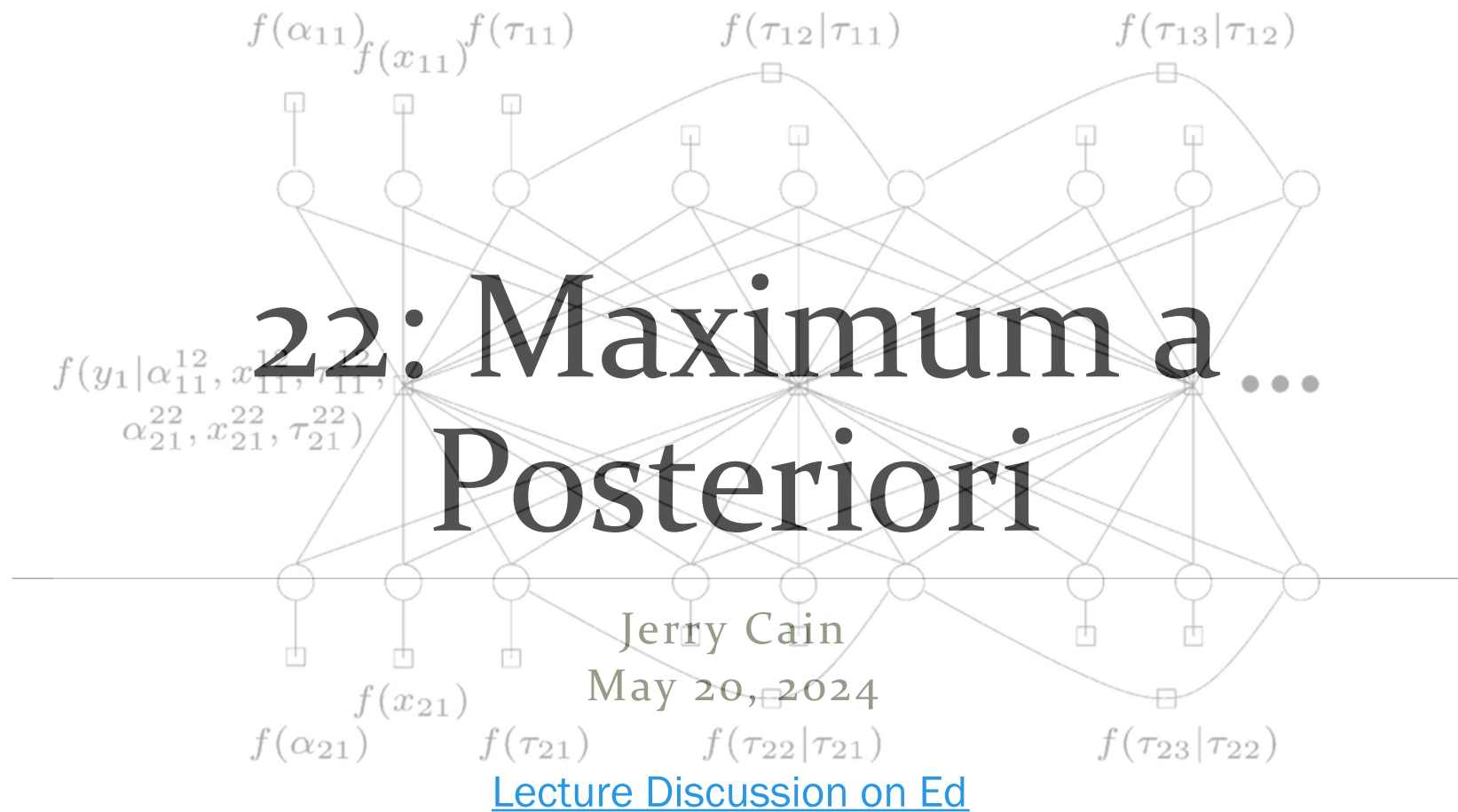
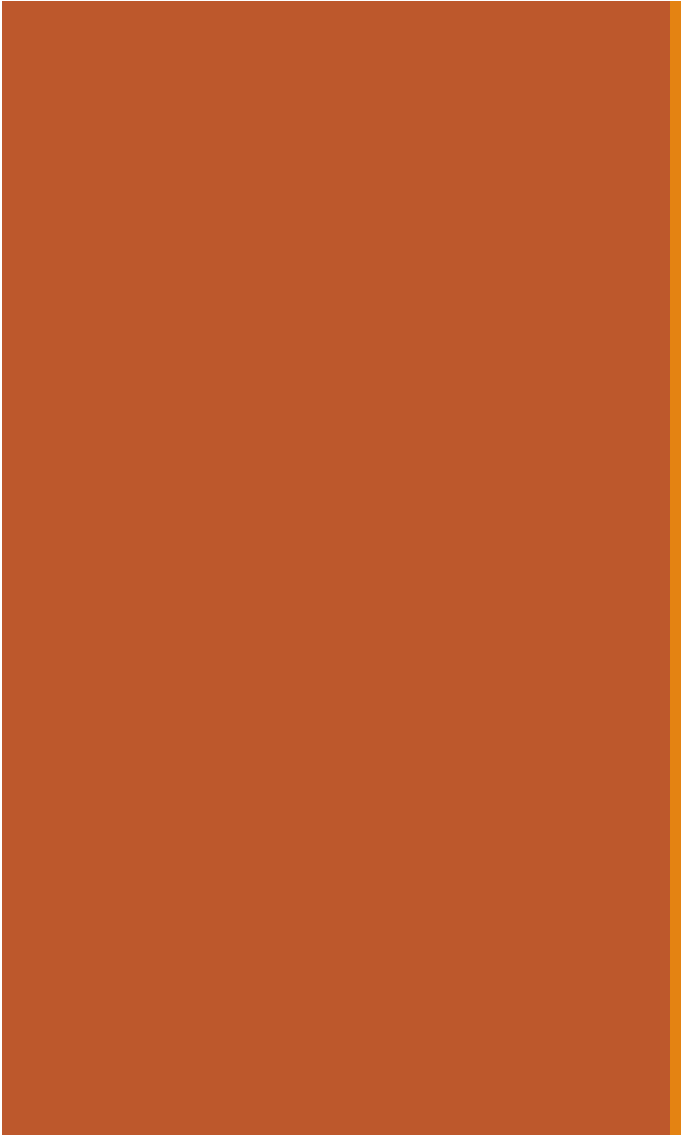






Maximum a Posteriori Estimator

Maximum A Posteriori (MAP) Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Maximum
Likelihood
Estimator
(MLE)

What parameter θ
maximizes the likelihood
of our observed data
($X_1, X_2, \dots, X_n$)?

goal: make this as large as possible

$$L(\theta) = f(X_1, X_2, \dots, X_n | \theta)$$
$$= \prod_{i=1}^n f(X_i | \theta)$$
$$\theta_{MLE} = \arg \max_{\theta} \underbrace{f(X_1, X_2, \dots, X_n | \theta)}_{\text{likelihood of data}}$$

Maximum
a Posteriori
(MAP)
Estimator

Given the sample data
($X_1, X_2, \dots, X_n$),
what is the **most probable**
parameter θ ?

goal: make this as large as possible

$$\theta_{MAP} = \arg \max_{\theta} \underbrace{f(\theta | X_1, X_2, \dots, X_n)}_{\text{posterior distribution of } \theta}$$

Maximum A Posteriori (MAP) Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

def The **Maximum a Posteriori (MAP) Estimator** of θ is the value of θ that maximizes the **posterior** distribution of θ .

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

Intuition with Bayes' Theorem:

$L(\theta)$, probability of data
given parameter θ

notice that
both the prior
and the
posterior
have θ as
their
primary
variables.

After seeing
data, posterior
belief of θ

$$\text{posterior } P(\theta | \text{data}) = \frac{\text{likelihood } P(\text{data} | \theta) \text{ prior } P(\theta)}{P(\text{data})}$$

Before seeing data,
prior belief of θ

Solving for θ_{MAP}

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) = \arg \max_{\theta} \frac{f(X_1, X_2, \dots, X_n | \theta) g(\theta)}{h(X_1, X_2, \dots, X_n)} \quad (\text{Bayes' Theorem})$$

↑
intentionally generic at this point

$$= \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)} \quad (\text{independence})$$

$$= \arg \max_{\theta} g(\theta) \prod_{i=1}^n f(X_i | \theta) \quad (1/h(X_1, X_2, \dots, X_n) \text{ is a positive constant w.r.t. } \theta)$$

$$= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right)$$

θ_{MAP} : Interpretation 1

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) = \arg \max_{\theta} \frac{f(X_1, X_2, \dots, X_n | \theta) g(\theta)}{h(X_1, X_2, \dots, X_n)} \quad (\text{Bayes' Theorem})$$

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$$= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right)$$

θ_{MAP} maximizes
log prior + log-likelihood

θ_{MAP} : Interpretation 2

- Observe data: $X_1, X_2, \dots, X_n$, all iid
- Let likelihood be same as MLE: $f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$
- Let the prior distribution of θ be $g(\theta)$.

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n) g(\theta)$$

The **mode** of the
posterior distribution of θ

(Bayes' Theorem)

$$= \arg \max_{\theta} \frac{g(\theta) \prod_{i=1}^n f(X_i | \theta)}{h(X_1, X_2, \dots, X_n)}$$

(independence)

$$= \arg \max_{\theta} g(\theta) \prod_{i=1}^n f(X_i | \theta)$$

($1/h(X_1, X_2, \dots, X_n)$ is a positive constant w.r.t. θ)

$$= \arg \max_{\theta} \left(\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta) \right)$$

θ_{MAP} maximizes
log prior + **log-likelihood**

Mode: A statistic of a random variable

The **mode** of a random variable X is defined as:

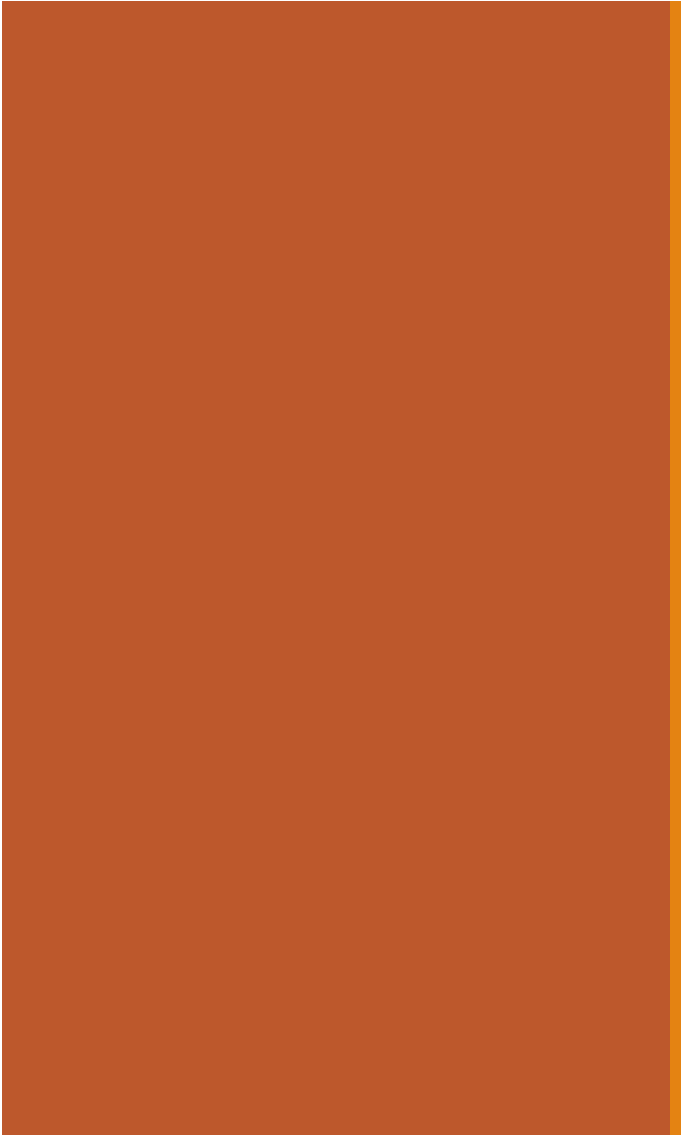
$$\begin{array}{ll} \arg \max_x p(x) & \arg \max_x f(x) \\ (X \text{ discrete, PMF } p(x)) & (X \text{ continuous, PDF } f(x)) \end{array}$$

- Intuitively: The value of X that is "most likely".
- Note some distributions don't have a unique mode (e.g., Uniform distribution, Bernoulli(0.5))

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

this is another maximization problem where we take derivatives and set them equal to 0.

θ_{MAP} is the most probable θ given the data $X_1, X_2, \dots, X_n$.



Bernoulli MAP: Choosing a prior

How does MAP work? (for Bernoulli)

Observe data

Choose model

Choose **prior on θ**

Find $\theta_{MAP} =$
 $\arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$

n heads, m tails

Bernoulli(p)

(some $g(\theta)$)

maximize
log prior + **log-likelihood**

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta)$$

- Differentiate, set to 0
- Solve

MAP depends on what
 $g(\theta)$ we choose.

MAP for Bernoulli

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior on θ . What is θ_{MAP} ?

perfectly reasonable!
it's centered at $p=0.5$, but
it allows for the actual
probability to be
slightly higher or
lower.

Suppose we pick a prior $\theta \sim \mathcal{N}(0.5, 1^2)$. $g(\theta) = \frac{1}{\sqrt{2\pi}} e^{-(p-0.5)^2/2}$

1. Determine log prior + log likelihood

$$\log g(\theta) + \log f(X_1, X_2, \dots, X_n | \theta)$$

$$= \log \left(\frac{1}{\sqrt{2\pi}} e^{-(p-0.5)^2/2} \right) + \log \left(\binom{n+m}{n} p^n (1-p)^m \right)$$

$$= -\log(\sqrt{2\pi}) - (p-0.5)^2/2 + \log \binom{n+m}{n} + n \log p + m \log(1-p)$$

2. Differentiate wrt (each) θ , set to 0

$$-(p-0.5) + \frac{n}{p} - \frac{m}{1-p} = 0$$

We should choose a prior that's easier to deal with. This one is hard!

3. Solve resulting equations

cubic equations, nope not going to do it

↑ reasonable question:
can we just choose a
prior because it's
easier?

A better approach: Use conjugate distributions

Observe data

Choose model

Choose **prior on θ**

Find $\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$

n heads, m tails

Bernoulli(p)

if the choice is really up to us, why not choose a prior that's easily manipulated

(some $g(\theta)$)

(choose conjugate distribution)

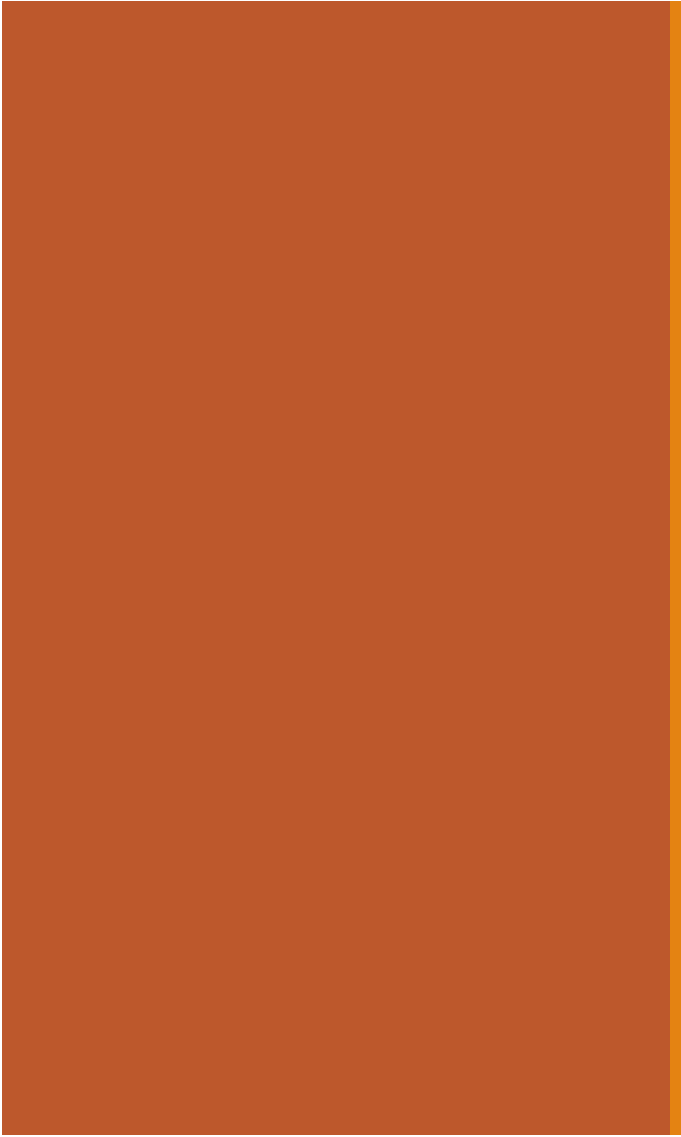
maximize
log prior + log-likelihood

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i | \theta)$$

- Differentiate, set to 0
- Solve



Up next: Conjugate priors are great for MAP!



Bernoulli MAP: Conjugate prior

Beta is a conjugate distribution for Bernoulli

Mostly Review

Beta is a **conjugate distribution** for Bernoulli, meaning:

- Prior and posterior parametric forms are the same
- Practically, conjugate means easy update:
Add numbers of "successes" and "failures" seen to Beta parameters.
- You can set the prior to reflect how fair/biased you think the experiment is a priori.

Beta(1,1) means you think all $\theta = p$ values equally likely.

Prior Beta($a = n_{imag} + 1, b = m_{imag} + 1$)

Experiment Observe n successes and m failures

Posterior Beta($a = n_{imag} + n + 1, b = m_{imag} + m + 1$)

Mode of Beta(a, b): $\frac{a - 1}{a + b - 2}$

(we'll prove this in a few minutes)

Beta parameters a, b are called **hyperparameters**.

Interpret Beta(a, b): $a + b - 2$ trials,
with $a - 1$ successes and with $b - 1$ failures

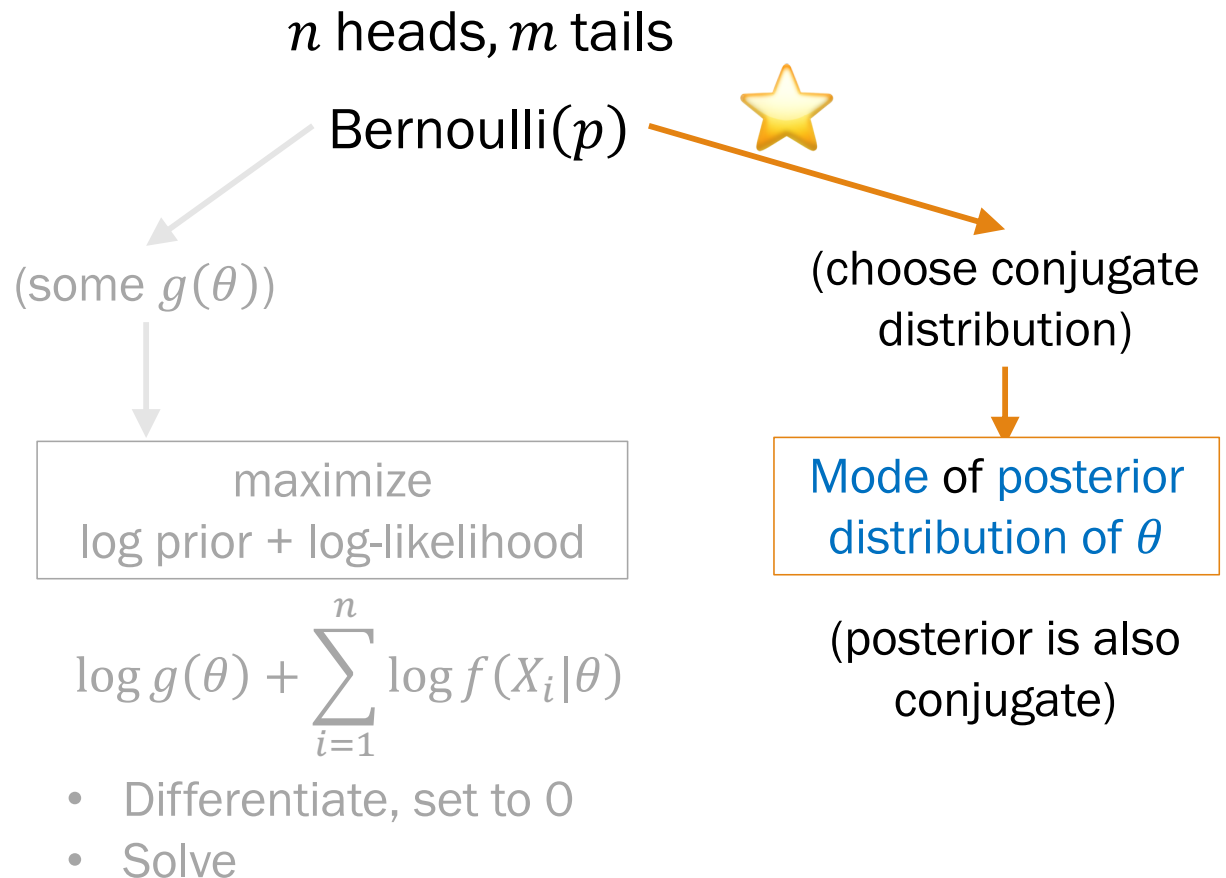
How does MAP work? (for Bernoulli)

Observe data

Choose model

Choose **prior on θ**

Find $\theta_{MAP} =$
 $\arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$



Conjugate strategy: MAP for Bernoulli

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail. } Define as data, D
- Choose a prior on θ . What is θ_{MAP} ?

1. Choose a prior

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$.

2. Determine posterior

Because Beta is a conjugate distribution for Bernoulli, the posterior distribution is $\theta|D \sim \text{Beta}(a + n, b + m)$

3. Compute MAP

$$\theta_{MAP} = \frac{a + n - 1}{a + n + b + m - 2} \quad (\text{mode of } \text{Beta}(a + n, b + m))$$



MAP in practice

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- What is the MAP estimator of the Bernoulli parameter p , if we assume a prior on p of Beta(2, 2)?



MAP in practice

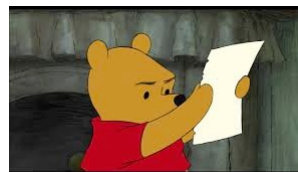
- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- What is the MAP estimator of the Bernoulli parameter p , if we assume a prior on p of Beta(2, 2)?

1. Choose a prior

$$\theta \sim \text{Beta}(2, 2).$$

mode of prior

$$\frac{2-1}{2+2-2} = \frac{1}{2}$$



Before flipping the coin,
we imagined 2 trials:
1 imaginary head, 1
imaginary tail.

2. Determine posterior

Posterior distribution of θ given observed data is Beta(9, 3)

9 is really 2 + 7
3 is really 2 + 1

3. Compute MAP

$$\theta_{MAP} = \frac{8}{10}$$

mode of posterior is $\frac{9-1}{9+3-2} = \frac{8}{10}$

After the experiment, we saw 10 trials:
8 heads (imaginary and real),
2 tails (imaginary and real).

Proving the mode of Beta

Observe data

Choose model

Choose prior on θ

Find $\theta_{MAP} = \arg \max_{\theta} f(\theta|X_1, X_2, \dots, X_n)$

These are **equivalent interpretations** of θ_{MAP} .

We'll use this equivalence to prove the mode of Beta.

n heads, m tails

Bernoulli(p)

(some arbitrary $g(\theta)$)

(choose conjugate)
Beta(a, b)

maximize
log prior + log-likelihood

**Mode of posterior
distribution of θ**

$$\log g(\theta) + \sum_{i=1}^n \log f(X_i|\theta)$$

(posterior is also conjugate)

- Differentiate, set to 0
- Solve

MAP for Bernoulli, conjugate prior (from first principles)

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior on θ . What is θ_{MAP} ?

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$. $g(\theta = p) = \frac{1}{\beta} p^{a-1} (1-p)^{b-1}$ normalizing
constant, β

1. Determine log prior + log likelihood

$$\begin{aligned} \log g(\theta) + \log f(X_1, X_2, \dots, X_n | \theta) &= \log \left(\frac{1}{\beta} p^{a-1} (1-p)^{b-1} \right) + \log \left(\binom{n+m}{n} p^n (1-p)^m \right) \\ &= \log \frac{1}{\beta} + \underbrace{(a-1) \log(p)} + \underbrace{(b-1) \log(1-p)} + \log \binom{n+m}{n} + \underbrace{n \log p} + \underbrace{m \log(1-p)} \end{aligned}$$

2. Differentiate w.r.t. (each) θ , set to 0

$$\underbrace{\frac{a-1}{p}} + \underbrace{\frac{n}{p}} - \underbrace{\frac{b-1}{1-p}} - \underbrace{\frac{m}{1-p}} = 0$$

3. Solve

(next slide)

MAP for Bernoulli, conjugate prior (from first principles)

- Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail.
- Choose a prior on θ . What is θ_{MAP} ?

Suppose we pick a prior $\theta \sim \text{Beta}(a, b)$. $g(\theta) = \frac{1}{\beta} p^{a-1} (1-p)^{b-1}$ normalizing
constant, β

3. Solve for p $\frac{a-1}{p} + \frac{n}{p} - \frac{b-1}{1-p} - \frac{m}{1-p} = 0$ (from previous slide)

$$\Rightarrow \frac{a+n-1}{p} - \frac{b+m-1}{1-p} = 0$$

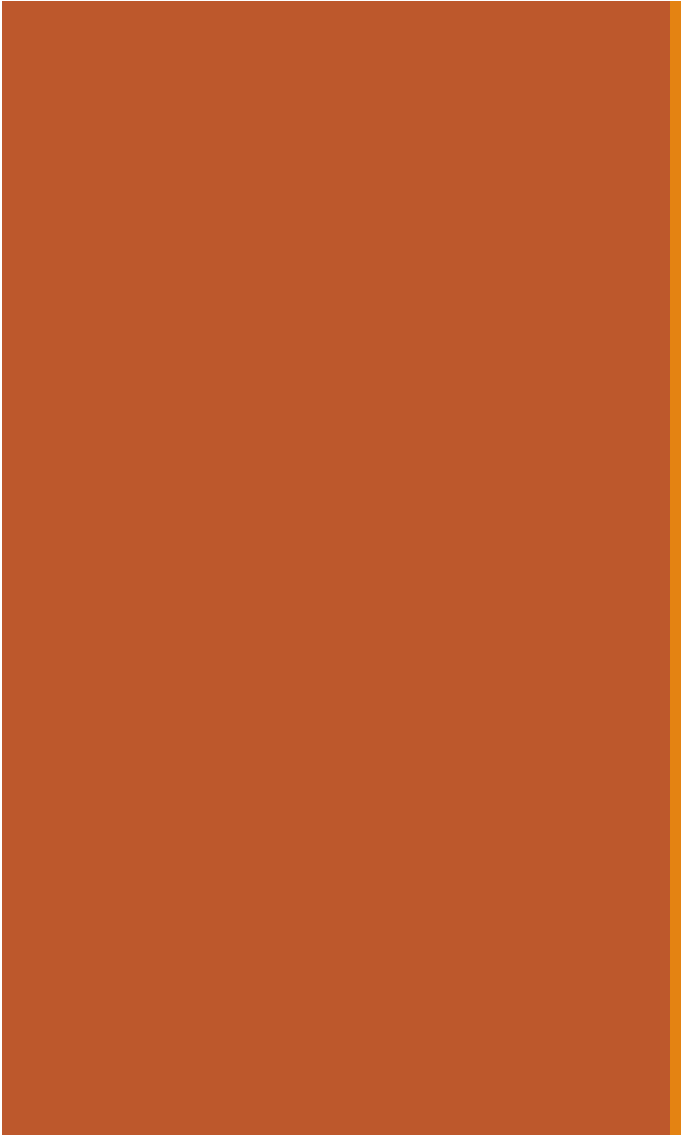
$$\Rightarrow (a+n-1) - (a+n-1)p = (b+m-1)p$$

$$\Rightarrow p(a+n+b+m-2) = a+n-1$$

$$\theta_{MAP} = \frac{a+n-1}{a+n+b+m-2} \quad \checkmark$$

The mode of the posterior,
 $\text{Beta}(a+n, b+m)$!

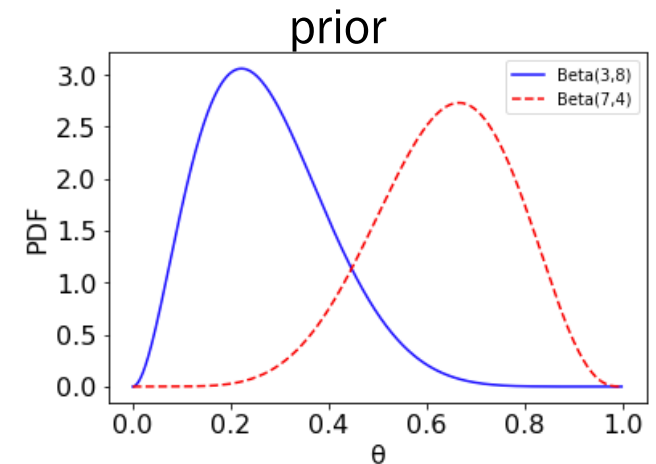
If we choose a conjugate prior, we avoid calculus with MAP, and we can simply report mode of posterior.



Choosing hyperparameters for conjugate prior

Where'd you get them priors?

- Let θ be the probability a coin turns up heads.
- Model θ with 2 different priors:
 - Prior 1: **Beta(3,8)**: 2 imaginary heads, 7 imaginary tails mode: $\frac{2}{9}$
 - Prior 2: **Beta(7,4)**: 6 imaginary heads, 3 imaginary tails mode: $\frac{6}{9}$

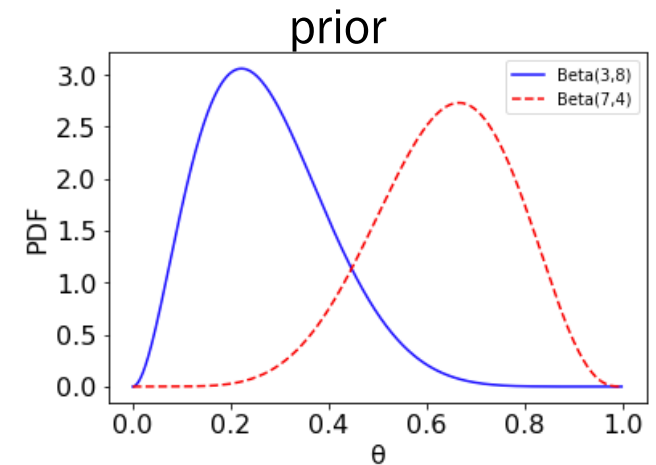


Now flip 100 coins and get 58 heads and 42 tails.

1. What are the two posterior distributions?
2. What are the modes of the two posterior distributions?

Where'd you get them priors?

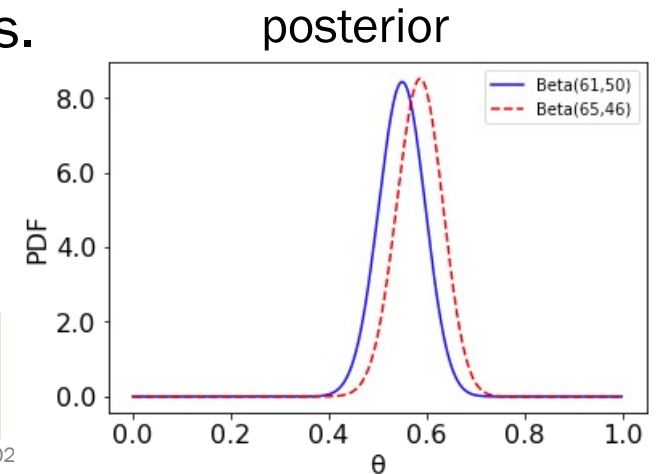
- Let θ be the probability a coin turns up heads.
- Model θ with 2 different priors:
 - Prior 1: **Beta(3,8)**: 2 imaginary heads, 7 imaginary tails mode: $\frac{2}{9}$
 - Prior 2: **Beta(7,4)**: 6 imaginary heads, 3 imaginary tails mode: $\frac{6}{9}$



Now flip 100 coins and get 58 heads and 42 tails.

Posterior 1: **Beta(61,50)** mode: $\frac{60}{109}$

Posterior 2: **Beta(65,46)** mode: $\frac{64}{109}$



Provided we collect enough data, posteriors will converge to the true value and choice of prior will matter less.

Laplace smoothing

MAP with **Laplace smoothing**: a prior which represents k imagined observations of each outcome.

- Categorical data (i.e., Multinomial, Bernoulli/Binomial)
- Also known as additive smoothing

Laplace estimate Imagine $k = 1$ of each outcome
(follows from Laplace's "[law of succession](#)")

Example: Laplace estimate for probabilities from previously mentioned experiment (100 coins: 58 heads, 42 tails)

$$\text{heads } \frac{59}{102} = \frac{58+1}{100+2} \quad \text{tails } \frac{43}{102} = \frac{42+1}{100+2}$$

Laplace smoothing:

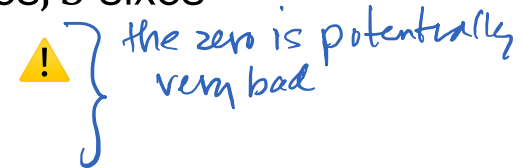
- Easy to implement/remember

Back to our happy Laplace

Consider our previous 6-sided die.

- Roll the dice $n = 12$ times.
- Observe: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

Recall θ_{MLE} :
 $p_1 = 3/12, p_2 = 2/12, p_3 = 0/12,$
 $p_4 = 3/12, p_5 = 1/12, p_6 = 3/12$



What are your Laplace estimates for each roll outcome?



Back to our happy Laplace

Consider our previous 6-sided die.

- Roll the dice $n = 12$ times.
- Observe: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

Recall θ_{MLE} : $p_1 = 3/12, p_2 = 2/12, p_3 = 0/12, \quad \text{!}$
 $p_4 = 3/12, p_5 = 1/12, p_6 = 3/12$

What are your Laplace estimates for each roll outcome?

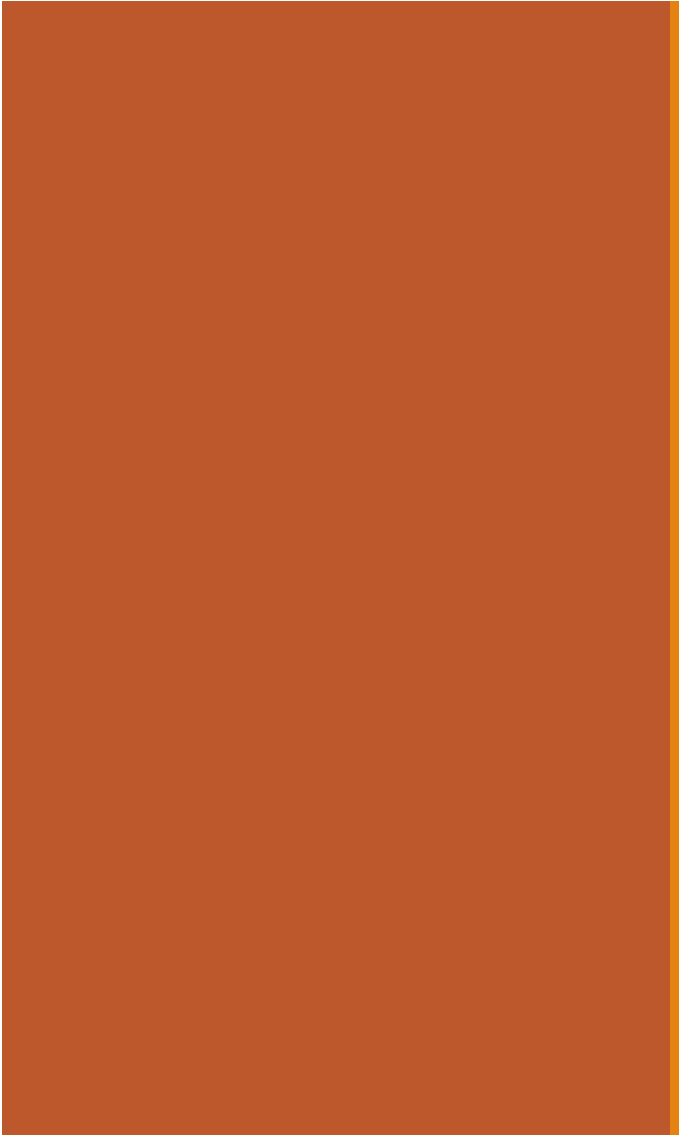
$$p_i = \frac{X_i + 1}{n + m}$$

$$X_3 = 0 \Rightarrow \frac{0 + 1}{12 + 6}$$

$$p_1 = 4/18, p_2 = 3/18, p_3 = 1/18, \quad \checkmark$$
$$p_4 = 4/18, p_5 = 2/18, p_6 = 4/18$$

Laplace smoothing:

- Easy to implement/remember
- **Avoids parameter estimation of 0**



Extra: Other Conjugates

Conjugate distributions

MAP
estimator:

$$\theta_{MAP} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

The **mode** of the
posterior distribution of θ

Distribution parameter	Conjugate distribution
Bernoulli p	Beta
Binomial p	Beta
Multinomial p_i	Dirichlet
Poisson λ	Gamma
Exponential λ	Gamma
Normal μ	Normal
Normal σ^2	Inverse Gamma

Multinomial is Multiple times the fun

Dirichlet($a_1, a_2, \dots, a_m$) is a conjugate for Multinomial.

- Generalizes Beta in the same way Multinomial generalizes Binomial:

$$f(x_1, x_2, \dots, x_m) = \frac{1}{B(a_1, a_2, \dots, a_m)} \prod_{i=1}^m x_i^{a_i-1}$$

Prior

Dirichlet($a_1, a_2, \dots, a_m$)

Saw $(\sum_{i=1}^m a_i) - m$ imaginary trials, with $a_i - 1$ of outcome i

Experiment Observe $n_1 + n_2 + \dots + n_m$ new trials, with n_i of outcome i

Posterior

Dirichlet($a_1 + n_1, a_2 + n_2, \dots, a_m + n_m$)

MAP:

$$p_i = \frac{a_i + n_i - 1}{(\sum_{i=1}^m a_i) + (\sum_{i=1}^m n_i) - m}$$

Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024



Good times with Gamma

Gamma(α, β) is a conjugate for Poisson.

- Also conjugate for Exponential, but we won't delve into that
- Mode of gamma: $(\alpha - 1)/\beta$

Prior

$$\theta \sim \text{Gamma}(\alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$$

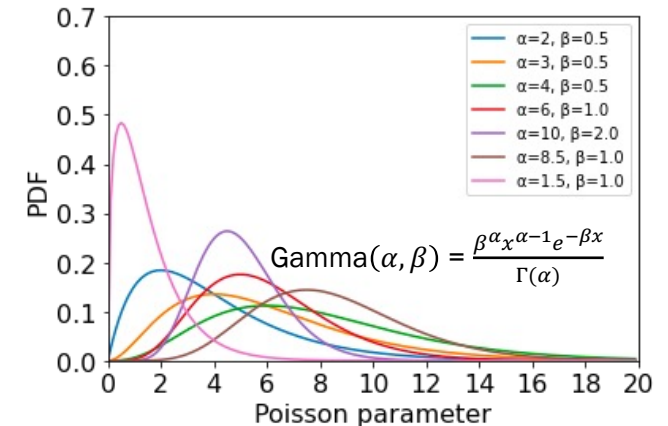
Saw $\alpha - 1$ total imaginary events during β prior time periods

Experiment Observe n events during next k time periods

Posterior $(\theta | n \text{ events in } k \text{ periods}) \sim \text{Gamma}(\alpha + n, \beta + k)$

MAP:

$$\theta_{MAP} = \frac{\alpha + n - 1}{\beta + k}$$



MAP for Poisson

Gamma(α, β)
is conjugate for Poisson Mode: $\frac{\alpha-1}{\beta}$

Let λ be the average # of successes in a time period.

1. What does it mean to have a prior of $\theta \sim \text{Gamma}(11, 5)$?

Observe 10 imaginary events
in 5 time periods,
i.e., observe at Poisson rate = 2

Now perform the experiment and see 11 events in next 2 time periods.

2. Given your prior, what is the posterior distribution?
3. What is θ_{MAP} ?



MAP for Poisson

Gamma(α, β)
is conjugate for Poisson Mode: $\frac{\alpha-1}{\beta}$

Let λ be the average # of successes in a time period.

1. What does it mean to have a prior of $\theta \sim \text{Gamma}(11, 5)$?

Observe 10 imaginary events
in 5 time periods,
i.e., observe at Poisson rate = 2

Now perform the experiment and see 11 events in next 2 time periods.

2. Given your prior, what is the posterior distribution?

$(\theta | n \text{ events in } k \text{ periods}) \sim \text{Gamma}(22, 7)$

3. What is θ_{MAP} ?

$\theta_{MAP} = 3$, the updated Poisson rate